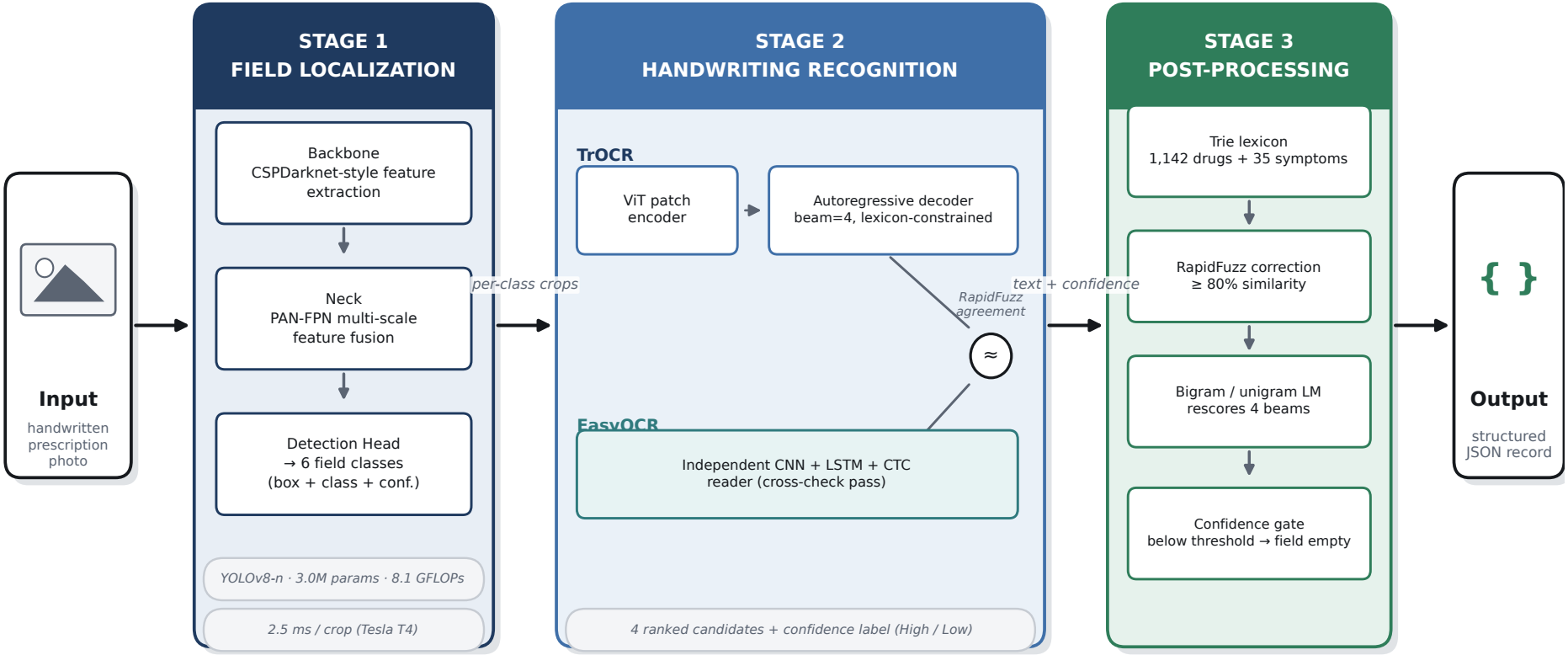
Input
handwritten prescription photo
STAGE 1
FIELD LOCALIZATION
Backbone
CSPDarknet-style feature extraction
Neck
PAN-FPN multi-scale feature fusion
Detection Head
→ 6 field classes
(box + class + conf.)
YOLOv8-n · 3.0M params · 8.1 GFLOPs
2.5 ms / crop (Tesla T4)
per-class crops
STAGE 2
HANDWRITING RECOGNITION
TrOCR
ViT patch encoder
Autoregressive decoder
beam=4, lexicon-constrained
RapidFuzz agreement
≈
EasyOCR
Independent CNN + LSTM + CTC
reader (cross-check pass)
4 ranked candidates + confidence label (High / Low)
text + confidence
STAGE 3
POST-PROCESSING
Trie lexicon
1,142 drugs + 35 symptoms
RapidFuzz correction
≥ 80% similarity
Bigram / unigram LM
rescores 4 beams
Confidence gate
below threshold → field empty
{ }
Output
structured JSON record